

Michael Medford, Ph.D

I am a staff-level software engineer and systems architect specializing in scalable data infrastructure, workflow orchestration, and operational automation for complex technical domains. My background in astrophysics and scientific computing has enabled me to design reliable systems that transform ambiguous, data-intensive workflows into scalable software platforms.

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Skills Python, Go, SQL, Distributed Systems, Workflow Orchestration, LLM Workflows, Retrieval-Augmented Workflows, Platform Architecture, PostgreSQL, Kafka, OpenSearch, Airflow, Prefect, Docker, Kubernetes, AWS, GCP, Operational Automation, Data Infrastructure, Cloud Architecture, Technical Leadership & Writing

Work Experience **Staff Software Engineer - Data Systems** KoBold Metals · April 2025 - Present

- Led organizational adoption of AI-assisted software development practices, including introducing local AI tooling workflows, redesigning engineering processes around increased development throughput, and mentoring engineers on AI-native development patterns using Cursor and agent-based workflows.
- Designed and operationalized the Korecam360 data infrastructure powering ingestion, synchronization, and retrieval of large-scale core imaging datasets across KoBold's exploration platform, enabling machine learning workflows, historical backfills, and downstream visualization systems.
- Built the Drillshift Daily Updates application after observing operational bottlenecks during fieldwork in Zambia, reducing drill superintendent reporting workflows from multiple hours each morning to approximately 15 minutes through automated workflow coordination and real-time operational reporting.

Staff Software Engineer - Subject Matter Expert in Datastores Insight M · July 2024 - March 2025

- Developed a two-year organizational roadmap to improve data architecture and security, including a staging and development data strategy, a data transformation pipeline for enhanced business intelligence dashboards, and a data retention overhaul, saving the company six figures annually
- Optimized database processes by refactoring high-cost computational components, reducing runtime expenses, and enhancing system resiliency with comprehensive test coverage
- Enhanced data infrastructure by implementing a database event alert pipeline, launching an internal application for query plan analysis to improve query efficiency, and introducing the company's first postmortem template to foster learning from software retrospectives

Staff Software Engineer - Systems Base Aumni, a JPMorgan Chase Company · July 2023 - June 2024

- Redesigned our backend architecture with management, product and engineering to enable flow in stream-aligned teams
- Developed a migration plan to maintain 100% uptime while refactoring our Ruby on Rails monolith into a set of independent services within the JPMorgan ecosystem
- Constructed contract testing framework to ensure independent deployability in collaboration with our Testing Enablement Team

Senior Software Engineer - Data Chapter Aumni, a JPMorgan Chase Company · November 2022 - June 2023

- Led and executed refactor of company's data processing model to unlock cross-functional analytics
- Designed, documented and deployed to production an independent microservice for indexing millions of documents into OpenSearch in only 6 weeks
- Invented an SQL unit test framework on top of a custom file parsing library that runs in Snowflake and executes via Airflow pipelines validating our core business logic
- Overhauled the infrastructure and tooling for developer data synchronization to reduce sync time from 15 minutes to 20 seconds

Technical Lead of Planet Fusion Monitoring Pod

Planet Labs · March 2022 - October 2022

- Architected 10x increase in our daily Python processing while cutting per unit costs 80% in four months
- 98% reduction in database query latency & CPU utilization via SQL schema & query optimization
- Cut GCS & AWS data delivery processing overhead by 30x using batched GCP pubsub messages
- Moved team into a bi-weekly sprint structure with ticket triage, sprint retrospectives and sprint planning

Geospatial Software Engineer

Planet Labs · June 2021 - October 2022

- Implemented automated data backups & disaster recovery via terraform to comply with ISO certification
- Set up on-call notifications by integrating Sentry, PagerDuty and Slack into our Python & Go stack
- Created a comprehensive cost estimate method to measure profit margin goals for our product teams

A New Method for Detecting Solar System Objects on High Performance Computers

Lawrence Berkeley National Laboratory · January 2016 - September 2019

w/ Dr. Peter Nugent

- Invented planet detection pipeline that searched 100+TB of images using 20,000+ lines of Python and C
- Implemented real-time neural network scoring of planet candidates in Dockerized Python Flask apps
- Engineered HPC scheduler to execute 1,000+ compute processes via many-to-many SQL databases
- Measured accuracy and completeness by applying statistical methods to artificially injected signals

Parallelized Executable for Removing Noise from Telescope Images

Lawrence Berkeley National Laboratory · September 2019 - June 2020

w/ Dr. Peter Nugent

- Constructed physical models of atmospheric fringes in optical images with principle component analysis
- Built parallelized feature identification and extraction tool currently running on 50,000+ images per night
- Released code as an open source Python package: *fringe* [fringe Documentation](#)

PhD, Astrophysics, University of California: Berkeley

May 2021

Advisors: Jessica Lu and Peter Nugent

Thesis: *Discovery of Rare Signals in Large Scale Time Domain Surveys: Dark Planets and Black Holes*

MA, Astrophysics, University of California: Berkeley

GPA: 3.79 / 4.00 | 2017

BS, Physics and Astronomy, Northwestern University

GPA: 3.75 / 4.00 | 2011

Weinberg College of Arts and Sciences

Advisor: Dr. Michael Smutko, Collaborator: Dr. Vicky Kalogera

BS, Theatre, Northwestern University

2011

School of Communication

PopSyCLE (2020)

[PopSyCLE Documentation](#)

Population Synthesis for Compact-object Lensing Events

- Developed pipeline infrastructure to execute code in high performance supercomputing environments
- Reconfigured data format schema to extendible compound HDF5 to enabling additional image filters

zort (2019)

[zort Documentation](#)

ZTF (Zwicky Transient Facility) Object Reader Tool

Sole Author

- Executed spatial cross-matching and filtering for time-domain measurements of billions of objects
- Official data reader for ZTF telescope Public Data Releases, representing 11+ international institutions

UC-National Lab In-Resident Graduate Fellowship

The Regents of the University of California

04/2019 - 03/2021

- Awarded \$130,000 to discover isolated black holes at the Lawrence Livermore National Laboratory

Astrophotography, Racquetball, Chess, Theatre, Financial Coaching